

3RD HARSH-ENVIRONMENT MASS SPECTROMETRY WORKSHOP

Multisensor Data Integration and Adaptive Sampling Strategies for an Autonomous Underwater Mass Spectrometer

Authors: Richard Camilli, Harold F. Hemond

Institute: Massachusetts Institute of Technology

ABSTRACT

Marine environments can present the challenges of vast spatial extent and highly dynamic environmental states. These factors, combined with the limited communication bandwidth usually available to a maneuvering underwater platform will often necessitate that autonomous decision and control processes be employed when studying these partially observable systems. We examine the concept of using real-time chemical data from an underwater mass spectrometer operating onboard an autonomous underwater vehicle (AUV), in concert with data from other sensors commonly used on these robots, for the purpose of pattern recognition and for providing effective adaptive sampling. This strategy is implicitly useful for a host of applications including concurrent and feature-based mapping, improving acoustic and magnetic sensor data interpretation, direction of chemotactic searches, and adaptive control of mass spectrometer scanning. A pattern recognition system within a layered control architecture is described for use with the NEREUS underwater mass spectrometer and the Odyssey autonomous underwater vehicle.